Quadratic Completion of Admissible Spectral Pairs via Binary-Icosahedral Representation

O26 — Spectral Admissibility Sub-programme

Jérôme Beau

Independent Researcher

April 2026

Abstract

We investigate the structural nature of the pair observable $\sigma_{\text{pair}}(n) = \sigma_c(n)\,\sigma_{q-c}(n)$ arising in the Weil-block analysis of the Heisenberg Cayley graphs $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$. Building on the canonical construction of conjugate pairs enforced by the parity involution [1] and the normalisation invariance established in O17–O19 [2, 1, 3], we construct an explicit dictionary between conjugate Weil blocks and rank-one matrix coefficients in a representation space.

We show that, in the pre-saturation regime, $\sigma_{\text{pair}}(n)$ has the same growth exponent as the Hilbert–Schmidt norm of an associated matrix trajectory, establishing a Level I identification (proved). We then formulate a hierarchy of stronger identifications: a quotient identification modulo normalisation (Level II), and a canonical representation-theoretic identification (Level III), which is a theorem conditional on a single structural hypothesis (the admissible embedding $\Phi_{q,\rho}$) in an isotypic sector of the binary icosahedral group $2I$ along the admissibility thread $Q_8 \subset 2I \subset \mathrm{SU}(2)$.

We provide concrete falsifiability tests based on the effective dimension of the trajectory in $\mathrm{End}(V_\rho)$, directly computable from O25 data [4]. A positive result would identify σ_{pair} as the restriction of a canonical Hermitian quadratic form and provide the first representation-theoretic interpretation of the admissible cascade exponent β^*. A negative result would still determine the correct ambient representation sector, refining the admissibility hierarchy without invalidating the dictionary.

Keywords. spectral admissibility; pair observable; Hilbert–Schmidt norm; Weil representation; conjugate pairs; representation theory; non-injective projection

Contents

1 Background and motivation

The spectral admissibility programme (O1–O25) studies the emergence of observable scaling laws from the non-injective projection structure of Cosmochrony [5], in which the static substrate χ is projected onto observable spacetime via Π. A central object is the Gram–Schmidt redundancy observable

$$\sigma_c(n) = \frac{\delta r_n}{|S_n|}, \tag{1}$$

defined on BFS shells S_n of Cayley graphs associated with $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$.

Early analyses based on scalar observables led to systematic discrepancies in the extracted exponent δ (O11–O14 [6, 7, 8, 9]). This mismatch was resolved in O16 [10] by identifying the correct observable as the *conjugate pair observable*

$$\sigma^{\mathrm{can}}_{\mathrm{pair}}(n) = \sigma_c(n)\,\sigma_{q-c}(n), \tag{2}$$

which doubles the exponent and yields values in the admissible window $\delta_{\mathrm{pair}} \in [7.4, 10.6]$.

The necessity of pairing is structural. O18 [1] shows that every fibre of the projection Π contains the parity involution $\chi \mapsto -\chi$, which induces the conjugation relation $\rho_{q-c} = \overline{\rho_c}$ at the Weil level. O17–O19 [2, 1, 3] establish that the apparent asymmetry between c and $q - c$ is a normalisation artefact, not a structural feature.

The present note addresses the following question:

> Is the bilinear form $\sigma^{\mathrm{can}}_{\mathrm{pair}}(n)$ an intrinsic quadratic object, or merely a product enforced by symmetry?

We show that $\sigma^{\mathrm{can}}_{\mathrm{pair}}$ admits a natural interpretation as the pullback of a Hermitian quadratic form on a representation space, and formulate precise criteria under which this interpretation becomes canonical.

2 The admissible $2I$ thread and the 600-cell

The admissibility analysis of [11, 12] classifies the finite subgroups of SU(2) compatible with bounded-flux admissibility constraints, identifying the chain

$$Q_8 \subset 2I \subset \mathrm{SU}(2), \tag{3}$$

where $2I$ is the binary icosahedral group of order 120. Among the admissible families, the binary polyhedral groups $2O$ and $2I$ are distinguished by the isotropy of their second-moment tensor [12], making $2I$ the maximal admissible finite subgroup of SU(2).

The group $2I$ admits irreducible representations of dimensions

$$d_\rho \in \{1, 2, 2, 3, 3, 4, 4, 5, 6\}, \tag{4}$$

and its matrix coefficient spaces $\mathrm{End}(V_\rho)$ have dimensions d_ρ^2, forming the canonical Peter–Weyl decomposition of $L^2(2I)$.

Geometrically, $2I$ is the rotational symmetry group of the regular 600-cell in $\mathbb{R}^4$, whose 120 vertices correspond to the elements of $2I$. The Laplacian spectrum of this structure can be organised representation-theoretically, with multiplicity patterns consistent with the Peter–Weyl dimensions d_ρ^2 of the irreps of $2I$ [11]. This is not a coincidence imported from external geometry: the same representation-theoretic hierarchy governs both the spectral decomposition of $2I$ and the organisation of admissible degrees of freedom along the thread $Q_8 \subset 2I \subset \mathrm{SU}(2)$.

The question addressed in this note is whether the conjugate-pair observable $\sigma^{\mathrm{can}}_{\mathrm{pair}}$ selects one of these isotypic sectors and realises its canonical quadratic form through the Gram–Schmidt dynamics.

3 The dictionary: conjugate Weil blocks and isotypic sectors of $2I$

This section establishes the explicit correspondence between the pair observable $\sigma^{\mathrm{can}}_{\mathrm{pair}}$ on conjugate Weil blocks $\{c, q-c\}$ and a natural Hermitian quadratic form on an isotypic sector of $2I$. The dictionary has three layers: the Weil side (§3.1), the $2I$ side (§3.2), and the identification map (§3.3).

3.1 The Weil side: structure of $\sigma^{\mathrm{can}}_{\mathrm{pair}}$

Fix a prime q and a generic block $c_{\mathrm{block}} = (c_1, c_2, c_3)$ with $c_i \neq 0$ and $c_1 + c_2 + c_3 \not\equiv 0 \pmod q$. The Gram–Schmidt redundancy at BFS shell n is

$$\sigma_c(n) = \frac{\delta r_n}{|S_n|}, \tag{5}$$

where δr_n counts the number of Weil fingerprint vectors in shell S_n linearly independent of the span built over shells $0, \ldots, n-1$ [7]. The canonical pair observable is

$$\sigma^{\mathrm{can}}_{\mathrm{pair}}(n) = \sigma_c(n)\, \sigma_{q-c}(n). \tag{6}$$

The structural basis of this definition rests on two results from O17–O18. First, the conjugation identity $\rho_{q-c} = \overline{\rho_c}$ together with the norm-invariance of Gram–Schmidt orthogonalisation implies $\delta_{q-c} = \delta_c$ for all admissible c [2]. Second, the proportionality $\sigma_{q-c}(n) = r(c,q)\, \sigma_c(n)$, with $r(c,q) \in \{1,2,3,4\}$ independent of n, holds to machine precision ($|\sigma_{q-c}(n)/\sigma_c(n) - r(c,q)| < 10^{-14}$ for all tested pairs and shells) [10]. Critically, O17 [2] demonstrates that $r(c,q)$ is a normalisation artefact of the pipeline determined by the initial supports (b_1, b_2) of each block sample, not an intrinsic invariant of the representation. The structurally robust quantity is therefore $\delta_c = \delta_{q-c}$, and the canonical form of the pair observable removes the artefact by working with the product $\sigma_c(n)\, \sigma_{q-c}(n)$ rather than each factor in isolation.

Remark 3.1 (Gauge-like nature of $r(c,q)$). The factor $r(c,q)$ plays the role of a gauge degree of freedom in the pipeline: it depends on sampling initial conditions, not on the representation. The matrix M_n defined in §3.3 must be constructed so that $\|\widetilde{M_n}\|^2_{\mathrm{HS}}$ is independent of this gauge choice, i.e. it absorbs $r(c,q)$ into a normalisation convention rather than treating it as structural data.

3.2 The $2I$ side: isotypic sectors and the Peter–Weyl quadratic form

For each irrep ρ of $2I$ with representation space V_ρ of dimension d_ρ, the Peter–Weyl theorem provides a canonical space of matrix coefficients:

$$\mathcal{H}_\rho = \{f : 2I \to \mathbb{C} \mid f(g) = \langle u, \rho(g)v\rangle,\ u, v \in V_\rho\} \simeq V_\rho \otimes \overline{V_\rho} \simeq \mathrm{End}(V_\rho), \tag{7}$$

which has dimension d_ρ^2. This space carries a canonical Hermitian inner product, the Hilbert–Schmidt form:

$$\langle A, B\rangle_{\mathrm{HS}} = \mathrm{tr}(A^\dagger B), \qquad \|A\|_{\mathrm{HS}}^2 = \mathrm{tr}(A^\dagger A), \quad A, B \in \mathrm{End}(V_\rho). \tag{8}$$

For a rank-one element $A = u \otimes \overline{v}$, this reduces to

$$\|u \otimes \overline{v}\|_{\mathrm{HS}}^2 = \|u\|^2 \cdot \|v\|^2. \tag{9}$$

This is the key structural fact: the Hilbert–Schmidt norm of a rank-one endomorphism is the product of the norms of its two factors.

3.3 The dictionary map

Construction. For each BFS shell n and each conjugate pair $\{c, q-c\}$, define the *un-normalised matrix coefficient*

$$\widetilde{M}_n := v_c^{(n)} \otimes \overline{v_{q-c}^{(n)}}, \tag{10}$$

where $v_c^{(n)}$ is the vector whose s-th component is the residual norm $\|r_s^{(n)}\|$ of the s-th Weil fingerprint vector in shell S_n after projection onto the Gram–Schmidt span built over shells $0, \ldots, n-1$, and similarly for $v_{q-c}^{(n)}$.

Central identity. By (9),

$$\|\widetilde{M}_n\|_{\mathrm{HS}}^2 = \|v_c^{(n)}\|^2 \cdot \|v_{q-c}^{(n)}\|^2 \sim \sigma_c(n) \cdot \sigma_{q-c}(n) = \sigma_{\mathrm{pair}}^{\mathrm{can}}(n). \tag{11}$$

Remark 3.2 (From pipeline quantity to structural norm). The proportionality $\|v_c^{(n)}\|^2 \sim \sigma_c(n)$ in (11) is not an additional assumption; it follows from the definition of the Gram–Schmidt observable and a single normalisation choice.

With the convention above, $\|v_c^{(n)}\|^2 = \sum_{s \in S_n} \|r_s^{(n)}\|^2$, the total squared residual energy contributed by shell S_n. By construction, $\sigma_c(n) = \delta r_n / |S_n|$, where δr_n is the rank increment of shell S_n.

Under the canonical normalisation of the fingerprint vectors adopted in O12–O19 [7, 2, 1, 3], each fingerprint vector has unit initial norm. The proportionality

$$\|v_c^{(n)}\|^2 = \kappa \cdot |S_n| \cdot \sigma_c(n) + \varepsilon_n \tag{12}$$

holds with $\kappa > 0$ a normalisation constant independent of n, and ε_n a remainder bounded by the contribution of vectors below the rank threshold. In the pre-saturation regime (the fitting window $[n_0, n_1]$ used throughout O16–O25), $\varepsilon_n / (|S_n| \cdot \sigma_c(n)) \to 0$ as the rank-threshold vectors become negligible relative to the total shell size.

Identity (11) therefore holds in the pre-saturation regime with a proportionality constant $\kappa^2 |S_n|^2$ absorbed into the pipeline normalisation. The power-law exponent δ_{pair}, extracted from the log-log regression of $\sigma_{\mathrm{pair}}^{\mathrm{can}}(n)$, is unaffected by this constant factor.

Three verification criteria.

(i) **Involution compatibility.** The parity involution $c \leftrightarrow q-c$ corresponds, via the conjugation identity $\rho_{q-c} = \overline{\rho_c}$, to complex conjugation $v \mapsto \overline{v}$ on the representation space. This is precisely the anti-linear involution that exchanges $u \otimes \overline{v}$ and $v \otimes \overline{u}$ in $\mathrm{End}(V_\rho)$, i.e. Hermitian adjunction $A \mapsto A^\dagger$. The pair observable $\sigma_{\mathrm{pair}}^{\mathrm{can}}(n) = \sigma_c(n)\,\sigma_{q-c}(n)$ is symmetric under this exchange, consistent with $\|A\|_{\mathrm{HS}}^2 = \|A^\dagger\|_{\mathrm{HS}}^2$. Involution compatibility holds.

(ii) **Gauge invariance.** The observable $\sigma_{\mathrm{pair}}^{\mathrm{can}}(n)$ is invariant under rescaling $v_c^{(n)} \mapsto \lambda v_c^{(n)}$, $v_{q-c}^{(n)} \mapsto \lambda^{-1} v_{q-c}^{(n)}$ with $\lambda > 0$, because $\sigma_c(n)$ scales as λ^2 and $\sigma_{q-c}(n)$ as λ^{-2}, leaving the product unchanged. Correspondingly, $\|\widetilde{M}_n\|_{\mathrm{HS}}^2$ is invariant under $(u, v) \mapsto (\lambda u, \lambda^{-1} v)$ in the rank-one element $u \otimes \overline{v}$. Gauge invariance holds.

(iii) δ_{pair} **stability.** The exponent δ_{pair} is extracted from the log-log regression of $\sigma_{\mathrm{pair}}^{\mathrm{can}}(n)$ over the fitting window. By (11), this is the same as the log-log regression of $\|\widetilde{M}_n\|_{\mathrm{HS}}^2$. The verticality theorem of O24 [4] establishes that δ_{pair} is stable under fibre perturbations; this translates, via the dictionary, into the statement that $\|\widetilde{M}_n\|_{\mathrm{HS}}^2$ grows with a well-defined power-law exponent independent of the choice of conjugate pair. δ_{pair} stability holds.

3.4 Strength of the identification

Identity (11) establishes the correspondence at *intermediate strength* in the sense of §4: $\sigma_{\mathrm{pair}}^{\mathrm{can}}(n)$ is proportional to $\|\widetilde{M}_n\|_{\mathrm{HS}}^2$, with a proportionality constant that is pipeline-normalisation-dependent and hence not intrinsic.

The *strong* identification requires the proportionality constant to be canonically fixed by the $2I$-representation theory, i.e. that $\widetilde{M}_n$ takes values in a specific isotypic component $\mathcal{H}_\rho$ for a representation ρ selected by the admissibility thread $Q_8 \subset 2I \subset \mathrm{SU}(2)$. The admissibility hierarchy of [11] selects the spin-$\frac{1}{2}$ sector (dimension $d_\rho = 2$, Laplacian eigenvalue $\lambda_{1/2} = 18$) as the dominant admissible sector on the $2I$ Cayley graph. If $v_c^{(n)} \in V_\rho$ for this sector, then $\widetilde{M}_n \in \mathrm{End}(V_\rho)$ has $d_\rho^2 = 4$ components. This identification is at present a conjecture, formulated precisely in §4.

4 The conjecture at three levels

Building on the dictionary of §3, we formulate a graduated identification between the pair observable $\sigma_{\mathrm{pair}}^{\mathrm{can}}(n)$ and the Hilbert–Schmidt quadratic form on an isotypic sector of $2I$. The three levels differ in the strength of the claim: growth equivalence (proved), quotient identification (structural), and canonical identification (conjectural).

4.1 Level I: Growth equivalence (proved)

Proposition 4.1 (Level I). *In the pre-saturation regime $n \in [n_0, n_1]$, the pair observable and the Hilbert–Schmidt norm of $\widetilde{M}_n$ have identical power-law growth exponents:*

$$\delta_{\mathrm{pair}} = \delta_{\mathrm{HS}}, \tag{13}$$

where δ_{HS} is defined by the log-log regression of $\|\widetilde{M}_n\|_{\mathrm{HS}}^2$ over the same fitting window.

Proof. By (11) and (12), $\sigma_{\text{pair}}^{\text{can}}(n) = \kappa^2 |S_n|^2 \|\widetilde{M}_n\|_{\text{HS}}^2 + O(\varepsilon_n)$, where the prefactor $\kappa^2|S_n|^2$ is independent of the conjugate pair. Taking log-log regression over $[n_0, n_1]$ and noting that $|S_n|$ grows as a fixed power of n (determined by the BFS geometry of G_q, independent of c), the regressions of $\log \sigma_{\text{pair}}^{\text{can}}$ and $\log \|\widetilde{M}_n\|_{\text{HS}}^2$ differ only in their intercepts. Hence $\delta_{\text{pair}} = \delta_{\text{HS}}$. $\square$

Remark 4.2. Level I is not a conjecture but a direct consequence of the dictionary construction of §3. It is stated separately to clarify what is proved and what remains conjectural at Levels II and III.

4.2 Level II: Quotient identification (structural)

The proportionality factor $\kappa^2|S_n|^2$ has two sources: the global normalisation convention of the fingerprint vectors (absorbed in κ) and the shell-size scaling (absorbed in $|S_n|^2$). Both are pipeline conventions rather than representation-theoretic data. Define the *normalisation equivalence*

$$A \approx B \quad \Longleftrightarrow \quad A = C \cdot B, \tag{14}$$

where $C > 0$ depends only on pipeline conventions and the BFS geometry of G_q, but not on the conjugate pair $\{c, q - c\}$ or the shell index n.

Conjecture 4.3 (Level II). *In the normalisation equivalence class defined above,*

$$\sigma_{\text{pair}}^{\text{can}}(n) \approx \|\widetilde{M}_n\|_{\text{HS}}^2. \tag{15}$$

Equivalently, $\sigma_{\text{pair}}^{\text{can}}(n)$ is the pullback of the Hilbert–Schmidt quadratic form on $\operatorname{End}(V_\rho)$ *under the dictionary map* (10)*, modulo a constant depending only on pipeline conventions.*

Level II would follow from Level I together with a proof that the proportionality constant $\kappa^2|S_n|^2$ does not depend on n in a way that could alter the pair-level structure. The n-dependence of $|S_n|^2$ is BFS-geometric and uniform across pairs, so the primary open point is the n-independence of κ.

4.3 Level III: Conditional theorem

The strong form requires the proportionality constant to be fixed canonically by the representation theory of $2I$. We show that this identification is a *theorem*, conditional on a single structural hypothesis: the existence of a canonical assignment from admissible Weil-block residuals to the spin-$\frac{1}{2}$ representation space of $2I$.

Hypothesis 4.4 (Canonical admissible embedding). *Let $\mathcal{V}_q$ denote the space generated by admissible Weil-block residual vectors $\{v_c^{(n)}\}$, equipped with the parity involution $c \mapsto q - c$, and let ρ be the spin-$\frac{1}{2}$ irreducible representation of $2I$ with representation space V_ρ (dimension $d_\rho = 2$).*

There exists a canonical linear map

$$\Phi_{q,\rho} : \mathcal{V}_q \longrightarrow V_\rho \tag{16}$$

satisfying:

(a) ***Involution equivariance.*** *The parity involution on $\mathcal{V}_q$ maps to the canonical anti-linear involution on V_ρ induced by Hermitian conjugation in* $\mathrm{End}(V_\rho)$*:*

$$\Phi_{q,\rho}(v_{q-c}^{(n)}) = \overline{\Phi_{q,\rho}(v_c^{(n)})}. \tag{17}$$

(b) ***Quadratic compatibility.*** *The pullback of the Hilbert–Schmidt form satisfies*

$$\|\Phi_{q,\rho}(v_c^{(n)})\|^2 = d_\rho^{-1}\,\|v_c^{(n)}\|^2 \quad \textit{for all } n \in [n_0, n_1]. \tag{18}$$

(c) ***Naturality.*** *The assignment $\Phi_{q,\rho}$ is canonical with respect to the admissible structure: it is independent of the choice of pair $\{c, q-c\}$ and of the shell index n, and depends only on (q, ρ).*

This assignment can be viewed as a natural morphism from the space of admissible residual configurations to an isotypic component of $2I$.

Hypothesis 4.4 isolates the only non-constructive step of the argument: all subsequent statements are formal consequences of the constructions of §3.

Theorem 4.5 (Level III: canonical quadratic realisation)**.** *Assume Hypothesis 4.4. Then for every admissible conjugate pair $\{c, q-c\}$ and every $n \in [n_0, n_1]$,*

$$\sigma_{\mathrm{pair}}^{\mathrm{can}}(n) = C_n \cdot \left\|\Phi_{q,\rho}(v_c^{(n)}) \otimes \overline{\Phi_{q,\rho}(v_{q-c}^{(n)})}\right\|_{\mathrm{HS}}^2, \tag{19}$$

where $C_n = \kappa^{-2}\,|S_n|^{-2}$ depends only on the pipeline normalisation and the BFS geometry, and not on the pair $\{c, q-c\}$.

In particular, $\sigma_{\mathrm{pair}}^{\mathrm{can}}(n)$ is the pullback of the canonical Hilbert–Schmidt quadratic form on $\mathrm{End}(V_\rho)$ *under $v_c^{(n)} \mapsto \Phi_{q,\rho}(v_c^{(n)})$, and therefore defines a rank-one trajectory in* $\mathrm{End}(V_\rho)$ *whose growth exponent coincides with δ_{pair}.*

Proof. Set $u^{(n)} := \Phi_{q,\rho}(v_c^{(n)})$ and $w^{(n)} := \Phi_{q,\rho}(v_{q-c}^{(n)})$. By condition (a), $w^{(n)} = \overline{u^{(n)}}$. By condition (b), $\|u^{(n)}\|^2 = d_\rho^{-1}\|v_c^{(n)}\|^2$ and $\|w^{(n)}\|^2 = d_\rho^{-1}\|v_{q-c}^{(n)}\|^2$. By (9),

$$\|u^{(n)} \otimes \overline{w^{(n)}}\|_{\mathrm{HS}}^2 = \|u^{(n)}\|^2 \cdot \|w^{(n)}\|^2 = d_\rho^{-2}\,\|v_c^{(n)}\|^2 \cdot \|v_{q-c}^{(n)}\|^2. \tag{20}$$

By (12), in the pre-saturation regime, $\|v_c^{(n)}\|^2 \cdot \|v_{q-c}^{(n)}\|^2 = \kappa^2\,|S_n|^2\,\sigma_c(n)\,\sigma_{q-c}(n) + O(\varepsilon_n)$. The factor d_ρ^{-2} is absorbed by setting $C_n = d_\rho^2\,\kappa^{-2}|S_n|^{-2}$, which gives (19). The growth exponent of $C_n\|\cdot\|_{\mathrm{HS}}^2$ equals that of $\sigma_{\mathrm{pair}}^{\mathrm{can}}$ by the same argument as Proposition 4.1. □

Remark 4.6. The constant C_n is uniform across pairs and does not affect the power-law exponent. The structural content of Theorem 4.5 is therefore entirely carried by the embedding $\Phi_{q,\rho}$. Hypothesis 4.4 has been resolved unconditionally in O27 [13]: the quaternionic rigidity theorem therein shows that any morphism satisfying conditions (a)–(c) necessarily factors through the admissible quotient $\mathcal{H}_{\mathrm{eff}} \simeq \mathfrak{su}(2)$, and that the canonical construction $\Phi_{q,\rho} = \rho \circ \iota \circ \pi$ is the unique such morphism up to unitary equivalence. In particular, naturality (c) is not an additional assumption but a structural consequence of conditions (a)–(b) and the Born–Infeld admissibility. Test 4 of §5 provides direct numerical evidence for the sector selected by this construction.

4.4 Candidate sector

The admissibility hierarchy of [11] selects the spin-$\frac{1}{2}$ sector of $2I$ as the minimal non-abelian admissible sector: dimension $d_\rho = 2$, Laplacian eigenvalue $\lambda_{1/2} = 18$ on the standard $2I$ Cayley graph. Under Hypothesis 4.4, the proportionality constant in Theorem 4.5 is $d_\rho^{-2} = \frac{1}{4}$, and $\Phi_{q,\rho}(v_c^{(n)}) \otimes \overline{\Phi_{q,\rho}(v_{q-c}^{(n)})}$ takes values in the 4-dimensional space $\mathrm{End}(V_\rho) \simeq \mathbb{C}^{2\times 2}$.

4.5 Structural interpretation

At Level I, σ_{pair} and the Hilbert–Schmidt norm share a growth exponent: the identification is asymptotic. At Level II, σ_{pair} is the Hilbert–Schmidt norm up to a normalisation: the identification is structural modulo convention. At Level III, σ_{pair} *is* the Hilbert–Schmidt norm squared of a rank-one element of $\mathrm{End}(V_\rho)$, with the normalisation fixed by representation theory: this is a theorem under Hypothesis 4.4, not a conjecture.

5 Discriminating criteria and falsifiability

The identification proposed in §4 leads to concrete, computable tests distinguishing the three levels. All tests require no modification of the O25 pipeline and can be evaluated directly from the stored Gram–Schmidt data.

5.1 Test 1: Involution compatibility

At the scalar level, the conjugation identity $\rho_{q-c} = \overline{\rho_c}$ implies $\delta_{q-c} = \delta_c$ [2]. At the matrix level, the dictionary maps $c \leftrightarrow q-c$ to $\widetilde{M}_n \leftrightarrow \widetilde{M}_n^\dagger$ in $\mathrm{End}(V_\rho)$. This is an independent empirical prediction: if the identification is structural, the Frobenius distance

$$\Delta_n(c) := \left\| \widetilde{M}_n(c)^\dagger - \widetilde{M}_n(q-c) \right\|_{\mathrm{HS}} \tag{21}$$

must vanish up to numerical precision for all $n \in [n_0, n_1]$.

Criterion 5.1 (Involution)**.** $\Delta_n(c) < \varepsilon_{\text{mach}}$ for all tested pairs and shells.

A non-zero $\Delta_n(c)$ that is nonetheless small would indicate that the identification holds only approximately, corresponding to Level II. A systematic non-zero value would falsify Levels II and III simultaneously.

5.2 Test 2: Gauge invariance

The factor $r(c, q)$ is a normalisation artefact of the pipeline [2]. The dictionary absorbs it by working with $\widetilde{M}_n$ rather than with individual normalised vectors.

Criterion 5.2 (Gauge invariance)**.** Under the rescaling $v_c^{(n)} \mapsto \lambda\, v_c^{(n)}$, $v_{q-c}^{(n)} \mapsto \lambda^{-1} v_{q-c}^{(n)}$ for $\lambda > 0$, $\|\widetilde{M}_n\|_{\mathrm{HS}}^2$ is invariant.

This follows from (9) by direct computation: $\|\lambda u \otimes \overline{\lambda^{-1} v}\|_{\mathrm{HS}}^2 = \lambda^2 \|u\|^2 \cdot \lambda^{-2} \|v\|^2 = \|u\|^2 \cdot \|v\|^2$. Test 2 is a consistency check on the construction, expected to hold identically.

5.3 Test 3: Exponent stability

Level I (Proposition 4.1) predicts $\delta_{\text{pair}} = \delta_{\text{HS}}$.

Criterion 5.3 (Exponent stability)**.** For each prime $q \in \{29, 61, 101, 151\}$, compute

$$\delta_{\text{HS}}(q) = \text{slope}\big(\log \|\widetilde{M}_n\|_{\text{HS}}^2 \text{ vs } \log(n+1)\big) \tag{22}$$

over the window $[n_0, n_1]$ used in O25 [14] (Table 2 therein). The difference $|\delta_{\text{HS}}(q) - \delta_{\text{pair}}(q)|$ must lie within the regression standard error reported in O25.

This test is the numerical anchor for the entire construction.

5.4 Test 4: Effective dimension of the trajectory

This is the central falsifiability test for Level III. For each conjugate pair $\{c, q-c\}$, construct the empirical covariance operator

$$\mathcal{C}_c = \frac{1}{N} \sum_{n=n_0}^{n_1} \text{vec}(\widetilde{M}_n)\,\text{vec}(\widetilde{M}_n)^\dagger, \quad N = n_1 - n_0 + 1, \tag{23}$$

where vec flattens $\widetilde{M}_n$ into a column vector.

Criterion 5.4 (Effective dimension)**.** Let r_{eff} be the numerical rank of $\mathcal{C}_c$ (number of singular values above threshold $10^{-10} \cdot \sigma_1(\mathcal{C}_c)$).

- If $r_{\text{eff}} = 4$ for all pairs: consistent with the spin-$\frac{1}{2}$ sector candidate ($d_\rho = 2$, $\dim \text{End}(V_\rho) = 4$).
- If $r_{\text{eff}} = d_\rho^2$ for some other fixed d_ρ: Level III holds for a different irrep of $2I$; the candidate sector must be revised.
- If r_{eff} varies across pairs or primes: Level III fails; the identification holds at most at Level II.

Remark 5.5 (Reformulation for conjugate-pair data — O29)*.* The Born–Infeld parity constraint $\rho_{q-c} = \bar{\rho}_c$ (O18) forces every outer product $\widetilde{M}_n$ into the complex symmetric subspace $\text{Sym}(V_\rho, \mathbb{C}) \subsetneq \text{End}(V_\rho)$ [15]. Consequently the accessible rank from conjugate-pair data satisfies the *symmetric rank formula* $r_{\text{eff}} = d_\rho(d_\rho+1)/2$, not d_ρ^2. Criterion 5.4 must therefore be reformulated as:

- If $r_{\text{eff}} = d_\rho(d_\rho+1)/2$ for a fixed integer d_ρ, universally across all pairs and primes: Level III holds for the sector of dimension d_ρ.

The observed $r_{\text{eff}} = 3$ (established in O28 [16] and confirmed in O29 [15]) has the unique positive integer inversion $d_\rho = 2$, confirming the spin-$\frac{1}{2}$ sector.

Remark 5.6*.* The matrix $\widetilde{M}_n$ is reconstructed from the stored vectors $v_c^{(n)}$ and $v_{q-c}^{(n)}$ by outer product. The covariance (23) and its SVD scale as $O(Nd^4)$ with d the effective dimension; for $d = 2$ this is negligible. No additional BFS computation or Weil decomposition is required.

5.5 Test 5: Universality across pairs

Level III requires that all admissible pairs for a given q probe the same representation sector.

Criterion 5.7 (Universality)**.** The singular value spectra of $\mathcal{C}_c$ and $\mathcal{C}_{c'}$ must agree up to numerical fluctuations for all admissible pairs c, c' at the same prime q.

A pair-dependent spectrum would indicate that different pairs project onto different subspaces of $\mathrm{End}(V_\rho)$, which is incompatible with a canonical identification. The inter-pair concentration of δ_{pair} established in O25 (coefficient of variation dropping from 6.1% to 1.5% over $q \in \{29, 61, 101, 151\}$) provides indirect support for this test.

5.6 Decision table

Outcome	Interpretation
Tests 1–3 fail	Dictionary construction fails; §3 must be revised
Tests 1–3 pass, r_{eff} variable	Level I only
Tests 1–3 pass, r_{eff} fixed $\neq 4$	Level II; wrong sector candidate
Tests 1–4 pass ($r_{\text{eff}} = 4$), Test 5 fails	Level II with spin-$\frac{1}{2}$ sector
Tests 1–5 pass	Level III; Hypothesis 4.4 supported; Theorem 4.5 applies

The row "Tests 1–3 pass, r_{eff} fixed $\neq 4$" is structurally informative even as a negative result: it identifies the correct representation sector without confirming the spin-$\frac{1}{2}$ candidate, pointing to a specific revision of the admissibility hierarchy argument.

6 Implications

6.1 From bilinear observable to canonical norm

Under Hypothesis 4.4, Theorem 4.5 establishes that σ_{pair} is the Hilbert–Schmidt norm squared of a rank-one element of $\mathrm{End}(V_\rho)$ selected by the admissibility thread $Q_8 \subset 2I \subset \mathrm{SU}(2)$, with normalisation fixed by d_ρ alone. The emergence of the pair observable is a *quadratic completion*: the passage from a linear object $v_c^{(n)} \in V_\rho$ to a rank-one operator $\Phi_{q,\rho}(v_c^{(n)}) \otimes \overline{\Phi_{q,\rho}(v_{q-c}^{(n)})} \in \mathrm{End}(V_\rho)$ endowed with its natural norm.

6.2 Interpretation of the exponent β^*

The transfer chain $c_\chi \to \delta_{\text{pair}} \to \beta^*$ was established unconditionally in O24 [4], where β^* is defined via

$$\beta^* \approx \frac{1}{\delta_{\text{pair}} + \frac{1}{2}}. \tag{24}$$

Under Level III, δ_{pair} characterises the growth of a Hilbert–Schmidt norm along a trajectory in $\mathrm{End}(V_\rho)$, and β^* acquires a direct representation-theoretic meaning: it is the effective scaling exponent of norm growth in the minimal admissible non-abelian sector. This interpretation is made unconditional in O27 [13] (Corollary 6.1), where the rigidity theorem establishes that this non-abelian sector is necessarily $\mathfrak{su}(2)$ and not any other structure. This provides a structural interpretation of the phenomenological window $\beta^* \in (0.09, 0.13)$ [4] as a constraint on admissible trajectories in representation space, rather than as a purely numerical fit.

6.3 Selection of the admissible sector

The spin-$\frac{1}{2}$ sector emerges as a candidate not from external physical input, but from internal admissibility constraints [11]. Test 4 of §5.4 provides a direct verification of this selection. If confirmed, $\mathrm{End}(V_\rho) \simeq \mathbb{C}^{2\times 2}$ is the minimal ambient space for admissible spectral dynamics. If a different fixed dimension is observed, the admissibility hierarchy must be revised accordingly, but the dictionary of §3 remains valid.

6.4 Role of non-injectivity

The construction makes explicit how non-injectivity of the projection $\Pi : \chi \to \mathcal{O}$ induces a quadratic structure at the observable level. The pairing $\{c, q-c\}$ is not an arbitrary symmetrisation, but the minimal realisation of the parity involution $\chi \mapsto -\chi$ derived in O18 [1] from the Born–Infeld action. The Hermitian structure of $\mathrm{End}(V_\rho)$ is therefore a projection-induced structure: it does not exist at the level of the substrate χ, but emerges from the identification of conjugate fibres under Π.

6.5 Outlook

The results of this note isolate a concrete bridge between the Weil representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, the admissibility thread $Q_8 \subset 2I \subset \mathrm{SU}(2)$, and the observable exponent δ_{pair}. The tests of §5 can be executed on O25 data [14] without additional computation. A positive result at Level III would provide the first representation-theoretic realisation of the admissible cascade exponent. A negative result with a fixed effective dimension $r_{\mathrm{eff}} \neq 4$ would identify the correct ambient sector, refining the admissibility hierarchy without invalidating the dictionary.

Subsequent to this paper, O27 [13] resolved Hypothesis 4.4 unconditionally by proving that any morphism satisfying conditions (a)–(c) is forced to factor through $\mathfrak{su}(2)$, with canonical construction $\Phi_{q,\rho} = \rho \circ \iota \circ \pi$ unique up to unitary equivalence. The full implication chain is thereby closed at the structural level:

$$\text{emergence} \Rightarrow \text{non-injectivity} \Rightarrow \text{pair structure} \Rightarrow \text{quadratic form} \Rightarrow \mathfrak{su}(2).$$

The numerical identification of the correct irreducible sector from O25 data — the value of d_ρ realised by Test 4 — is completed in O29 [15]. The symmetric rank formula $r_{\mathrm{eff}} = d_\rho(d_\rho+1)/2$ yields $d_\rho = 2$ from the observed $r_{\mathrm{eff}} = 3$, confirming the spin-$\frac{1}{2}$ sector and closing the implication chain of O27 [13].

Remark 6.1 (Resolution of Test 4 in O29). Test 4 is resolved in O29 [15]. The key structural result is that the Born–Infeld parity constraint forces the outer products $\widetilde{M}_n$ into $\mathrm{Sym}(V_\rho, \mathbb{C})$ (see Remark 5.5), so that the correct target is $r_{\mathrm{eff}} = d_\rho(d_\rho+1)/2$ rather than d_ρ^2. Numerically, $r_{\mathrm{eff}} = 3$ holds universally for $q \in \{61, 151\}$ (100% of pairs, zero inter-pair variance) and is confirmed modally for $q = 101$ by a multi-sample analysis; the unique inversion gives $d_\rho = 2$ (spin-$\frac{1}{2}$).

References

[1] J. Beau. Minimal fibre structure of the non-injective projection from born–infeld indiscernability: Derivation of the parity involution, 2026. O18 paper.

[2] J. Beau. Fibre-level admissibility from conjugate weil blocks: Structural derivation of the pair observable, 2026. O17 paper.

[3] J. Beau. About canonical pair observables in weil blocks: Structure of the residual amplitude factor and pipeline-independent formulation, 2026. O19 paper.

[4] J. Beau. Vertical non-injectivity and the stability of the observable rank: Closing the unconditional transfer $c_\chi \to \delta_{\mathrm{pair}} \to \beta^*$, 2026. O24 paper.

[5] J. Beau. Cosmochrony: A pre-geometric relational framework for emergent physics, 2026. Cosmochrony white paper.

[6] J. Beau. Weil-block projective capacity on heisenberg graphs: Resolving the representation obstruction and extracting the capacity exponent, 2026. O11 paper.

[7] J. Beau. Exact weil-block projective capacity on heisenberg graphs: Resolving the final obstruction to δ extraction, 2026. O12 paper.

[8] J. Beau. Asymptotic stability of exact weil-block capacity on heisenberg graphs: Extended prime range, variance reduction, and requalification of the $\delta \to \beta^*$ tension, 2026. O13 paper.

[9] J. Beau. Observable-class mismatch and the corrected $\delta \mapsto \beta^*$ relation on heisenberg graphs: Block normalisation, central-phase contribution, 2026. O14 paper.

[10] J. Beau. Fibre admissibility and resolution of the capacity exponent discrepancy, 2026. O16 paper.

[11] J. Beau. Spectral admissibility under bounded flux: Non-abelian mode selection on cayley graphs of su(2) subgroups, 2026. O1 paper.

[12] J. Beau. Pairwise character geometry and the classification of neutral generators in su(2), 2026. O3 paper.

[13] J. Beau. Quaternionic rigidity of admissible morphisms: Every admissible $\phi_{q,\rho}$ necessarily factors through $\mathfrak{su}(2)$, 2026. O27 paper.

[14] J. Beau. Systematic pair-level campaign for δ_{pair}: Convergence, inter-pair concentration, and normalization structure, 2026. O25 paper.

[15] J. Beau. Spin-$\frac{1}{2}$ sector identification via the symmetric rank formula: Effective dimension of the admissible covariance in $\mathrm{End}(v_\rho)$, 2026. O29 paper.

[16] J. Beau. Asymptotic calibration of the bfs window and effective dimension of the admissible trajectory, 2026. O28 paper.